CLARA

Cardiovascular Clinical Question Bank

100 Selected Questions from the
NLM Clinical Questions Collection

Research and Development Evaluation Material

Version 1.0
2 November 2026

Prepared for the CLARA Computer Science thesis project.
Selected candidate questions for evaluating a retrieval-augmented generation assistant.

Answer relevancy: reviewer instructions

Read each predefined question alongside its corresponding CLARA-generated answer. Assign one score from 1 to 5 using the criteria below. Rate how directly and completely the answer addresses the question, and whether it includes unnecessary content.

Five-point answer relevancy scale

1: Does not address the question.

2: Mostly unrelated; addresses only a small part.

3: Partially addresses the question, with substantial omissions.

4: Directly addresses most of the question, with minor omissions.

5: Directly and fully addresses the question without unnecessary content.

How to record the assessment

Record the question number, relevance score, and a brief reason in the accompanying answer-review sheet. Flag clinical inaccuracies or safety concerns separately; a relevant answer is not necessarily medically correct.

This researcher-developed ordinal scale measures clinician-rated answer relevancy. It is separate from the automated RAGAS Answer Relevancy metric. The rubric should be reviewed before scoring; this document does not establish that it is a validated instrument.

Reviewer instructions added: 6 November 2026.

Document scope and source attribution

This document contains 100 unique cardiovascular questions selected from 481 cardiovascular-tagged records imported from the NLM Clinical Questions Collection. It is a CLARA project compilation, not an official NLM publication or a clinically validated examination.

The National Library of Medicine (NLM, 2025) describes the collection as questions gathered from healthcare providers in clinical settings between 1991 and 2003. The collection is historical and is no longer expanded with new content.

Selection and presentation

Selection emphasized adult cardiovascular diagnosis, treatment, monitoring, prevention, and medication safety, with reduced repetition. This purposive selection is not a statistical sample or a validated ranking of clinical importance.

Question text uses the source general_question field when available, otherwise the original_question field. Only display whitespace is normalized. Each item includes its NLM ID and original collection record ID. The accompanying JSONL retains the original wording and archive provenance.

Evaluation status

Reference answers and evaluation scores are not included. CLARA responses will be reviewed using the answer relevancy rubric on the preceding page. Reference answers must be prepared from the relevant guideline PDFs and reviewed separately. Validation remains pending until the reviewer completes the final sign-off. Historical question wording is retained for traceability.

Source citation

National Library of Medicine. (2025). *Clinical Questions Collection* [Data set]. Data.gov. https://catalog.data.gov/dataset/clinical-questions-collection

Source archive references and attribution details appear after Question 100.

Question bank overview

1. Hypertension: diagnosis, treatment, and monitoring

Questions 1-25 | 25 questions

2. Heart failure: assessment and treatment

Questions 26-40 | 15 questions

3. Lipids and cardiovascular prevention

Questions 41-50 | 10 questions

4. Coronary disease, chest pain, and cardiac testing

Questions 51-62 | 12 questions

5. Arrhythmias and conduction disorders

Questions 63-74 | 12 questions

6. Stroke, transient ischemic attack, and carotid disease

Questions 75-82 | 8 questions

7. Anticoagulation and thromboembolic disease

Questions 83-92 | 10 questions

8. Peripheral vascular disease and broader cardiovascular assessment

Questions 93-100 | 8 questions

Reading the source labels

For example, NQ000767 is the NLM question identifier and IA-0767 is the source record identifier within Collection IA. These identifiers support checking the selected wording against the downloaded XML records.

1. Hypertension: diagnosis, treatment, and monitoring

1. What is the best way to decide whether to go by home blood pressures that are normal versus office blood pressures that are high?

Source: NLM | NQ000767 | IA-0767 | Collection IA

2. How many elevated blood pressures are required to make the diagnosis of hypertension?

Source: NLM | NQ003926 | IC-0926 | Collection IC

3. What are the indications for a 24-hour blood pressure monitor?

Source: NLM | NQ001878 | IB-0895 | Collection IB

4. Does it make any difference if you take the blood pressure again in the same arm after a few seconds of the first reading? Do you get an accurate reading the second time?

Source: NLM | NQ001995 | IB-1012 | Collection IB

5. Does it make any difference whether someone has their shirt on with a rolled up sleeve when you take their blood pressure?

Source: NLM | NQ001994 | IB-1011 | Collection IB

6. What antihypertensives should be used according to what comorbid conditions are present?

Source: NLM | NQ002571 | IB-1621 | Collection IB

7. What two- and three-drug combinations are reasonable for hypertension?

Source: NLM | NQ002609 | IB-1659 | Collection IB

8. What is the drug of choice to treat hypertension in someone with chronic renal failure?

Source: NLM | NQ002167 | IB-1186 | Collection IB

9. What are good drugs to treat hypertension in the elderly?

Source: NLM | NQ002720 | IB-1775 | Collection IB

10. What antihypertensives are safe to use when the patient wants to get pregnant?

Source: NLM | NQ003606 | IC-0606 | Collection IC

11. What antihypertensives are safe while breast feeding?

Source: NLM | NQ001548 | IB-0562 | Collection IB

12. What does the workup for secondary causes of hypertension consist of?

Source: NLM | NQ002966 | IB-2114 | Collection IB

13. What are the options for ruling out renal artery stenosis in a patient with refractory hypertension?

Source: NLM | NQ002801 | IB-1857 | Collection IB

14. If a patient is on three medications for hypertension which is not controlled, should a fourth medication be added or should a substitution be made?

Source: NLM | NQ003776 | IC-0776 | Collection IC

15. What are effective techniques to aid medication compliance in patients with hypertension?

Source: NLM | NQ000047 | IA-0047 | Collection IA

16. What is the recommended approach to an angiotensin-converting-enzyme-inhibitor-induced cough in terms of evaluation and treatment?

Source: NLM | NQ000805 | IA-0805 | Collection IA

17. Do nonsteroidal anti-inflammatory drugs cause hypertension? If so, how commonly?

Source: NLM | NQ002714 | IB-1767 | Collection IB

18. Are oral decongestants contraindicated in patients with hypertension?

Source: NLM | NQ002103 | IB-1122 | Collection IB

19. Should a low diastolic blood pressure which causes an increase in the pulse pressure be treated?

Source: NLM | NQ003715 | IC-0715 | Collection IC

20. What testing and what treatment is indicated for a patient with orthostatic hypotension which may be exacerbating coronary artery spasm?

Source: NLM | NQ003275 | IC-0275 | Collection IC

21. What is the definition of malignant hypertension (criteria for diagnosis)?

Source: NLM | NQ004046 | IC-1046 | Collection IC

22. How do you use labetalol to acutely lower blood pressure?

Source: NLM | NQ002537 | IB-1586 | Collection IB

23. How often should patients with stable hypertension and hyperlipidemia be seen?

Source: NLM | NQ000696 | IA-0696 | Collection IA

24. What was the outcome of the controversy about whether sodium in the diet aggravates hypertension?

Source: NLM | NQ001612 | IB-0627 | Collection IB

25. What class of antihypertensives have the least effect on male sexual function?

Source: NLM | NQ000374 | IA-0374 | Collection IA

2. Heart failure: assessment and treatment

26. What are New York Heart Association classes?

Source: NLM | NQ000225 | IA-0225 | Collection IA

27. In a patient with congestive heart failure, how can you tell whether they have systolic or diastolic dysfunction?

Source: NLM | NQ001583 | IB-0597 | Collection IB

28. What is a normal left ventricular ejection fraction?

Source: NLM | NQ002170 | IB-1189 | Collection IB

29. What would cause global hypokinesis on an echocardiogram (ejection fraction 40%)? What kinds of cardiomyopathy are there?

Source: NLM | NQ002725 | IB-1780 | Collection IB

30. What is the best way to determine whether the recent onset of dyspnea in a patient with chronic congestive heart failure is due to the heart failure or something else?

Source: NLM | NQ001065 | IA-1065 | Collection IA

31. What drugs should patients with heart failure be on unless contraindicated, besides diuretics?

Source: NLM | NQ002726 | IB-1781 | Collection IB

32. When should you use carvedilol or other beta blockers (for example, metoprolol) for congestive heart failure?

Source: NLM | NQ002536 | IB-1585 | Collection IB

33. How do you treat congestive heart failure when you also have chronic renal failure?

Source: NLM | NQ001514 | IB-0528 | Collection IB

34. Are angiotensin converting enzyme inhibitors contraindicated in hypertensive patients with a creatinine over a certain level?

Source: NLM | NQ001731 | IB-0747 | Collection IB

35. How do you treat refractory congestive heart failure?

Source: NLM | NQ001494 | IB-0508 | Collection IB

36. How do you treat diastolic dysfunction when seen on echocardiogram?

Source: NLM | NQ002883 | IB-2030 | Collection IB

37. Is there any treatment that improves survival in patients with diastolic dysfunction?

Source: NLM | NQ003528 | IC-0528 | Collection IC

38. When should a potassium-sparing diuretic be used instead of potassium supplementation?

Source: NLM | NQ000376 | IA-0376 | Collection IA

39. In a patient with alcoholic cardiomyopathy, is it OK to drink any alcohol at all?

Source: NLM | NQ003413 | IC-0413 | Collection IC

40. What causes jugular venous distension besides congestive heart failure?

Source: NLM | NQ002439 | IB-1488 | Collection IB

3. Lipids and cardiovascular prevention

41. What is the best treatment for hypercholesterolemia in an elderly (mid-70's) patient?

Source: NLM | NQ000391 | IA-0391 | Collection IA

42. At what age should screening cholesterol testing be discontinued?

Source: NLM | NQ000368 | IA-0368 | Collection IA

43. In a patient status post coronary angioplasty, should elevated lipids be treated initially with diet only or should drugs be used as initial therapy?

Source: NLM | NQ001071 | IA-1071 | Collection IA

44. What is the recommended management for an elderly woman with elevated cholesterol (total cholesterol approximately 290, low-density-lipoprotein (LDL) cholesterol approximately 210)?

Source: NLM | NQ000850 | IA-0850 | Collection IA

45. If someone is taking lovastatin and they get sore muscles but have a normal CPK (creatine phosphokinase), do you still need to worry about myositis from lovastatin?

Source: NLM | NQ001950 | IB-0967 | Collection IB

46. How high can you let the creatine phosphokinase get in someone on Mevacor (lovastatin) before you have to stop the drug?

Source: NLM | NQ002082 | IB-1101 | Collection IB

47. What types of fat are good for preventing coronary disease?

Source: NLM | NQ002399 | IB-1445 | Collection IB

48. Who should receive primary prophylaxis against vascular events with aspirin?

Source: NLM | NQ003529 | IC-0529 | Collection IC

49. Should a young woman with a strong family history of coronary disease at a young age be screened for coronary disease?

Source: NLM | NQ003445 | IC-0445 | Collection IC

50. Is it okay to use the nicotine patch in someone with known coronary artery disease?

Source: NLM | NQ001751 | IB-0767 | Collection IB

4. Coronary disease, chest pain, and cardiac testing

51. What is the best way to determine whether chest pain is due to coronary disease?

Source: NLM | NQ000509 | IA-0509 | Collection IA

52. What are the indications for immediate evaluation of a patient who gives a history of angina?

Source: NLM | NQ000553 | IA-0553 | Collection IA

53. What are the possible causes of chest pain in women, and what is the best way to distinguish among the possibilities?

Source: NLM | NQ000587 | IA-0587 | Collection IA

54. How do you treat an acute myocardial infarction in the emergency room? What things should be done?

Source: NLM | NQ002808 | IB-1864 | Collection IB

55. Should patients who are status post myocardial infarction be placed on beta blockers, aspirin, and angiotensin-converting-enzyme inhibitors indefinitely?

Source: NLM | NQ002804 | IB-1860 | Collection IB

56. How do you interpret Troponin levels?

Source: NLM | NQ002187 | IB-1206 | Collection IB

57. What is the difference between ischemia, injury, and infarction on the electrocardiogram?

Source: NLM | NQ002286 | IB-1309 | Collection IB

58. What is the relative accuracy and cost of various stress tests for coronary artery disease?

Source: NLM | NQ003532 | IC-0532 | Collection IC

59. Why should you do a pharmacologic stress echocardiogram rather than an exercise stress echocardiogram in a patient with left bundle branch block?

Source: NLM | NQ003747 | IC-0747 | Collection IC

60. When should a patient who is status post coronary bypass surgery or other procedure for coronary artery disease have followup exercise testing?

Source: NLM | NQ003521 | IC-0521 | Collection IC

61. Should a patient with typical chest pain and a negative treadmill 3 years ago have further testing?

Source: NLM | NQ003019 | IC-0019 | Collection IC

62. What is the best cardiac evaluation to do in a patient following a coronary artery bypass graft, who has terrible lungs and can't do a treadmill?

Source: NLM | NQ000925 | IA-0925 | Collection IA

5. Arrhythmias and conduction disorders

63. What is the recommended management of new onset atrial fibrillation?

Source: NLM | NQ000847 | IA-0847 | Collection IA

64. What drug should be used for chemical cardioversion of atrial fibrillation?

Source: NLM | NQ001956 | IB-0973 | Collection IB

65. How do you differentiate ventricular tachycardia from supraventricular tachycardia with aberrancy?

Source: NLM | NQ002006 | IB-1023 | Collection IB

66. How do you treat ventricular tachycardia?

Source: NLM | NQ002917 | IB-2065 | Collection IB

67. What causes Torsades de Pointes and how do you treat it? Overdrive with external pacemaker?

Source: NLM | NQ001710 | IB-0726 | Collection IB

68. When should you be concerned about frequent premature ventricular contractions in an adult?

Source: NLM | NQ002455 | IB-1504 | Collection IB

69. What are the possible causes of syncope, and what is the best way to distinguish among the possibilities?

Source: NLM | NQ000505 | IA-0505 | Collection IA

70. Should a young woman with anxiety and palpitations have a Holter monitor?

Source: NLM | NQ003689 | IC-0689 | Collection IC

71. What should you do with extreme bradycardia (35 beats per minute) in an asymptomatic adult?

Source: NLM | NQ002681 | IB-1734 | Collection IB

72. What are the indications for a pacemaker in a patient with intermittent third-degree heart block?

Source: NLM | NQ000612 | IA-0612 | Collection IA

73. What are the side effects of amiodarone?

Source: NLM | NQ002192 | IB-1211 | Collection IB

74. Could verapamil, a beta blocker, and digoxin cause enough bradycardia to produce weakness and dizziness?

Source: NLM | NQ000319 | IA-0319 | Collection IA

6. Stroke, transient ischemic attack, and carotid disease

75. What is the recommended evaluation in a patient with transient ischemic attacks?

Source: NLM | NQ000032 | IA-0032 | Collection IA

76. In a patient with transient ischemic attacks but no heart murmur, is an echocardiogram indicated?

Source: NLM | NQ000033 | IA-0033 | Collection IA

77. In a patient with atrial fibrillation, what are the possible causes of a dizzy episode besides a transient ischemic attack, and what is the best way to distinguish among the possibilities?

Source: NLM | NQ000663 | IA-0663 | Collection IA

78. Is Plavix (clopidogrel) effective in preventing transient ischemic attacks? What are the indications?

Source: NLM | NQ002518 | IB-1567 | Collection IB

79. When does the maximal swelling and clinical deterioration occur after an intracerebral hemorrhage?

Source: NLM | NQ002477 | IB-1526 | Collection IB

80. What are the indications for carotid endarterectomy?

Source: NLM | NQ003423 | IC-0423 | Collection IC

81. What is the recommended evaluation and treatment of patients with a carotid bruit?

Source: NLM | NQ000089 | IA-0089 | Collection IA

82. What are the ultrasound (duplex doppler) recommendations for surveillance after carotid endarterectomy?

Source: NLM | NQ003366 | IC-0366 | Collection IC

7. Anticoagulation and thromboembolic disease

83. What are the indications for starting Coumadin in a patient with chronic atrial fibrillation?

Source: NLM | NQ000939 | IA-0939 | Collection IA

84. What is the best way to decide whether a patient with atrial fibrillation at high risk of falling should be on Coumadin?

Source: NLM | NQ000845 | IA-0845 | Collection IA

85. How should you anticoagulate someone in atrial fibrillation before you cardiovert them?

Source: NLM | NQ002078 | IB-1097 | Collection IB

86. How long should you keep someone anticoagulated after a successful cardioversion?

Source: NLM | NQ002064 | IB-1083 | Collection IB

87. How should you suspend anticoagulants for surgery? Should you use heparin, low-molecular weight heparin, or nothing and for how long?

Source: NLM | NQ002172 | IB-1191 | Collection IB

88. What would cause the protime to fluctuate from time to time on the same dose of Coumadin?

Source: NLM | NQ000903 | IA-0903 | Collection IA

89. How do you dose Vitamin K if you want to reverse the effects of warfarin because the prothrombin time is too long?

Source: NLM | NQ002608 | IB-1658 | Collection IB

90. What is the dose of Lovenox and do you need to alter the dose of the elderly or if there is renal insufficiency?

Source: NLM | NQ003302 | IC-0302 | Collection IC

91. What is the reference to that algorithm for diagnosing a pulmonary embolus and what is the strategy?

Source: NLM | NQ002242 | IB-1265 | Collection IB

92. How do you treat a deep-vein thrombosis and pulmonary embolus during pregnancy and how do you monitor the patient on heparin or low-molecular-weight heparin?

Source: NLM | NQ003308 | IC-0308 | Collection IC

8. Peripheral vascular disease and broader cardiovascular assessment

93. How can you distinguish the leg pain that comes from intermittent claudication as opposed to diabetic neuropathy?

Source: NLM | NQ000332 | IA-0332 | Collection IA

94. How do you interpret different ankle/brachial blood pressure indices? What's bad, what's normal, is there a cutoff ratio for considering surgery?

Source: NLM | NQ001111 | IB-0019 | Collection IB

95. What are the 5 P's of acute arterial occlusion?

Source: NLM | NQ002643 | IB-1694 | Collection IB

96. Is it reasonable to get periodic ultrasounds to rule out an abdominal aortic aneurysm in someone with a family history of abdominal aortic aneurysm?

Source: NLM | NQ002297 | IB-1320 | Collection IB

97. How often should a patient have an ultrasound or computerized tomography (CT) to follow a thoracic aortic aneurysm?

Source: NLM | NQ000715 | IA-0715 | Collection IA

98. What is the recommended followup for a patient with an incidentally noted Type B aortic dissection on chest computerized tomogram?

Source: NLM | NQ003455 | IC-0455 | Collection IC

99. Should asymptomatic patients get screening ultrasound evaluations for carotid artery stenosis, abdominal aortic aneurysms, or peripheral vascular disease?

Source: NLM | NQ003367 | IC-0367 | Collection IC

100. What are the indications for an echocardiogram in a young adult with a Grade I or II murmur?

Source: NLM | NQ000191 | IA-0191 | Collection IA

References and source attribution

National Library of Medicine. (2025). *Clinical Questions Collection* [Data set]. Data.gov. https://catalog.data.gov/dataset/clinical-questions-collection

National Library of Medicine. (n.d.). *IOWA Collection IA questions* [XML data set]. Clinical Questions Collection. https://data.lhncbc.nlm.nih.gov/public/Clinical-Questions/Collection-IA-1095-Questions.zip

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National Library of Medicine. (n.d.). *IOWA Collection IC questions* [XML data set]. Clinical Questions Collection. https://data.lhncbc.nlm.nih.gov/public/Clinical-Questions/Collection-IC-1062-Questions.zip

Attribution and traceability

All selected questions are drawn from Collections IA, IB, and IC. Collection ID was included in the candidate import but contributes no items to this shortlist. Item-level source labels identify the corresponding XML records. The 2025 date above refers to the catalog-reported dataset update, not the year each clinical question was asked.

The Data.gov catalog identifies the dataset license as the Open Data Commons Open Database License (ODbL). Source attribution is retained; the selection and organization are specific to the CLARA project. Catalog and download links were checked on 2 November 2026.

The citations identify the origins of the questions. They are not clinical evidence supporting an answer. References for eventual evaluation answers must point to the actual guideline evidence used.

Validation statement - complete after review

By signing below, I confirm that I have reviewed the submitted dataset, selected questions, and accompanying CLARA-generated answers for their stated research purpose. My validation is limited to the materials reviewed and is subject to my recorded comments and required corrections.

Validated by: Johnrey T. Pepito

Team Manager
New Marketlink Pharmaceutical Corporation
Health & Wellness Segment

Signature: _____ Date: _____